5.8 Phenotypic Variability

▼ What is osteogenesis imperfecta?

- An **inherited disorder** characterized by **extreme fragility of the bones.**
- **Osteogenesis imperfecta** is split into **nine subtypes.**
- It is also known as **brittle bone disease**

▼ What is muscular dystrophy?

- **Muscular dystrophies** are a **group of diseases** which **cause progressive weakness and muscle breakdown.**
- The most common type is **Duchenne muscular dystrophy (DMD)** which **accounts for approximately 50% of all cases**

▼ What causes muscular dystrophy?

- All of the **genes affected** are involved in **attaching dystrophin through the cell membrane to the collagen** in the **extracellular matrix** either **directly or in the post translation modification of the proteins involved.**

▼ What is the Multiple Endocrine Neoplasia type 1 (MEN1) gene?

- This is a disease which **increases the carriers' chance of developing adenomas in endocrine tissue**

▼ What is hereditary haemochromatosis?

- It is caused by a **mutation in the human homeostatic iron regulator protein (HFE)**
- This **affects the way in which dietary iron is absorbed** leading to **excess iron absorption.**
- **Lower levels of intake** are associated with **improved disease prognosis** in patients with Hereditary haemochomotosis.

▼ How can the presence of multiple genes have an impact on phenotypic expression of a particular disorder?

- Interactions between disease causing genetic mutations and other genes can drastically alter the phenotype displayed by patients with the same mutation.

▼ How can different mutations in the same gene can lead to different phenotypes?

- Some diseases are associated with only a **single mutation** for example sickle cell anemia. Where as **other disease** can arise as the result of **many mutations within the same gene.**
- There may also be variation in the type of mutation that occurs in the gene.

▼ What is a trinucleotide repeat disorder?

- A **mutation** in which a **region of three repeated nucleotides (i.e CAG for Hungtington's)** in the genome **increases in number during DNA replication.**
- If there are fewer than 27 repeats in the genome, these tend to be stable, and the function of the protein remains normal.
- However once the threshold is exceeded, proteins become defective and **noticeable changes in phenotype occur.**

▼ How can trinucleotide repeat expansions be transferred?

- These can be **transferred vertically**, where the trinucleotide repeat **increases in number** when passed down **from parent to offspring**

▼ What phenotypic effect is characteristic of an unstable trinucleotide repeat?

- **Earlier age of onset** in offspring

▼ How can we distinguish between differences in phenotypes caused by second gene interference and the environment?

- Differences in phenotypes caused by **second gene interference** is usually **more severe** (i.e the phenotypes are more dissimilar) than the **environment where the phenotypic differences are more subtle.**